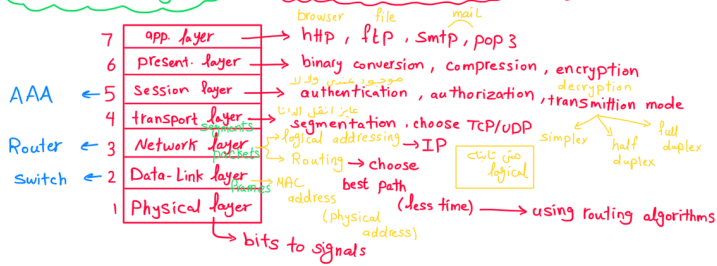


7  
6  
5  
4  
3  
2  
1  
All People Seem to Need Data Processing

## Ch:4 OSI Model (30% from previous final)

### Open System Interconnection 7 layers



TCP  
↓  
الدقة (وقت)  
checking regularly

UDP  
↓  
السرعة و خفيف

app. layer

- HTTP → browser (port 80) Hyper Text Transfer Protocol
- FTP → file < sftp File Transfer Protocol
- DNS → convert domain name to IP address Domain Name System (UDP port 53)  
↓  
hyperlink
- SMTP → Sending mails over TCP port 25 or 587. Simple Mail Transfer Protocol
- POP3 → receiving mails
- Telnet → remote session on a device TCP port 23  
→ replaced by SSH (more secured)

pres. layer

- SSL/TLS → encryption Secure Socket Layer  
Transport Layer Security

Session layer

- NetBIOS → name services
- PPTP → VPN protocol Point-to-Point Tunneling Protocol

transp. layer

- TCP → reliable delivery of data Transmission Control Protocol
- UDP → provide fast delivery of data User Datagram Protocol (but not efficient)

network layer

- IP → logical address Internet Protocol
- ARP → Maps IP to the mac address Address Resolution Protocol
- ICMP → detect errors Internet Control Message Protocol
- IPsec → IP more secured used in VPNs. Internet Protocol Security

data link layer

- PPP → □ □ direct connection between 2 nodes Point-to-Point Protocol
- ATM → transfer data Asynchronous Transfer Mode (replaced by ethernet)
- Ethernet → used in local networks

Physical layer

- USB → data transfer
- Bluetooth → short-range
- IEEE 802.11 (wifi)